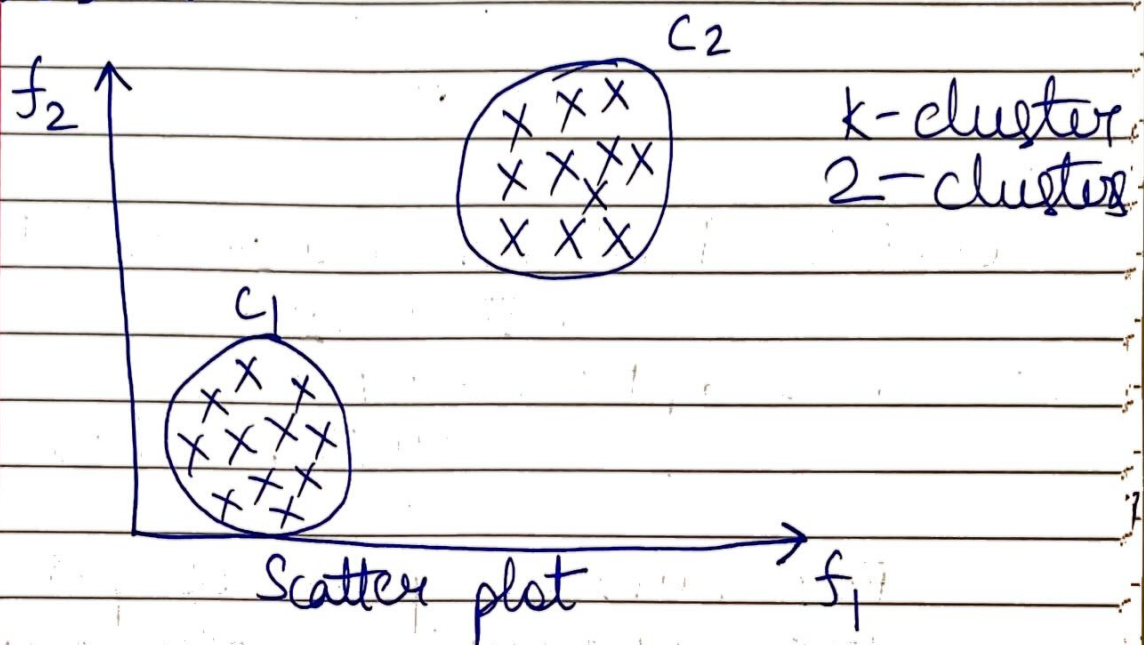


ch1 - Unsupervised learning / clustering

Video 4 Clustering Metrics

Geometrically, what is a clustering result:



$$D = \{x_i\}, x_i \in \mathbb{R}^2$$

Intra cluster (within a cluster)
Inter cluster (across or b/w cluster)

Cluster points in a group is such that intra cluster points have very less distance while inter cluster have large distance. This is the basis of how we measure clustering.

Ideally, inter-cluster distance should be very high while intra cluster distance is very low.

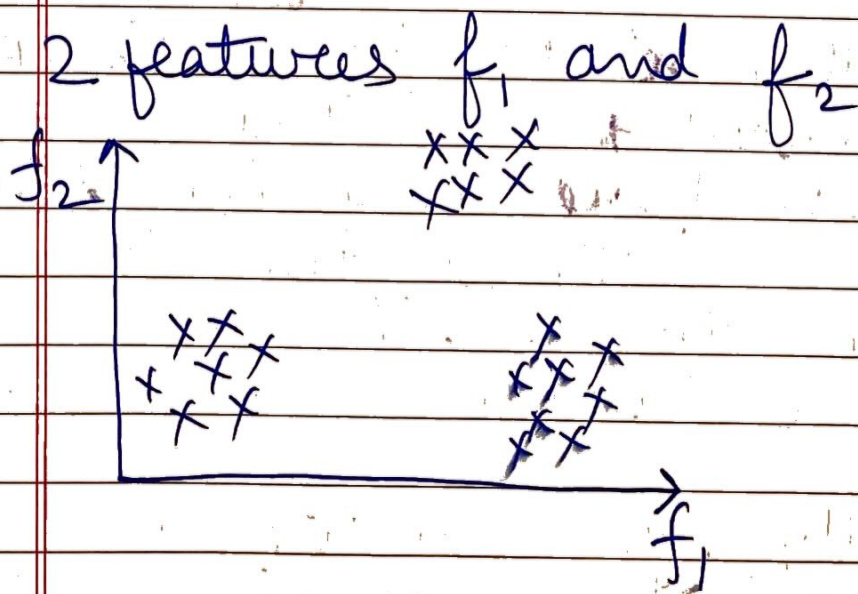
1) Dunn Index

$$D = \frac{\min_{i,j} d(i,j)}{\max_K d'(K)}$$

Should be $\uparrow$	Nunn - Minimal inter cluster distance
Should be $\downarrow$	Deno - Maximal intra cluster distance

For every pair of points take minimum of the two points in inter cluster

Video 5 K-means geometric intuition, Centroids



k - no. of clusters. It is hyperparameter in k -means clustering.

So, if $k=3$, there are 3 clusters.

clusters have a centroid.

C_1, C_2, C_3 are clusters ^{centroids} of S_1, S_2, S_3
sets of clusters. $S_1 \cup S_2 \cup S_3 = D$

$$S_1 \cap S_2 = \phi \text{ (Null set) (Dataset)}$$

$$S_1 \cap S_3 = \phi$$

$$S_2 \cap S_3 = \phi$$

You get k -centroids in k -Means clustering.

k -clusters	1	2	3	...	k
k -centroids	C_1	C_2	C_3	...	C_k
k -sets	S_1	S_2	S_3	...	S_k

$$C_i = \frac{1}{n} \sum_{x_j \in S_i} x_j \leftarrow \text{Mean point to } S_i$$

k -means is a centroid based clustering technique.

Big challenge in k-means clustering is:-

How to find k-centroids

If you get k-centroids
you get k sets from the
nearest centroid.

video 6 K-Means: Mathematical Formulation: objective function

For dataset $D = \{x_1, x_2, \dots, x_n\}$

Task Find k centroids: $c_1, c_2, \dots, c_k$
Sets $S_1, S_2, \dots, S_k$

For centroids $\forall x_i \in S_j$

$$\forall_{i,j} S_i \cap S_j = \emptyset$$

$$\text{argmin}_{c_1, c_2, \dots, c_k} \sum_{i=1}^k \sum_{x \in S_i} \|x - c_i\|^2$$

Squared distance of x from c_i

such that $x \in S_i$
 $S_i \cap S_j = \emptyset$

constraints

proximity

$$\begin{cases} C_1, C_2, \dots, C_k \\ S_1, S_2, \dots, S_k \end{cases}$$

Formula literally means
intra-cluster distance is minimized

Formula in English:

Find argmin of sum of all the clusters of sum of squared distance for centroid in cluster i such that $x_0 \in S_i$ and $S_i \cap S_j = \emptyset$

This problem is very very hard to solve. it is exponential in pts. $O(2^n)$

If problem is hard, solve it approximately.

Lloyd's algo is 1 such approx algo

Video 7 K-means algorithm

Lloyd's algorithm

1) Initialization

- Randomly pick k points from D (dataset) and call them $c_1, c_2, \dots, c_k$

2) Assignment

- For each point x_i in D select the nearest c_j
- Compute $\text{dist}(x_i, c_j) \forall j = 1, 2, \dots, k$
- Add x_i to Set S_j

3) Recompute centroid

- Recalculate/update c_j 's as follows:

$$c_j = \frac{1}{|S_j|} \sum_{x_i \in S_j} x_i$$

$$\forall c_1, c_2, \dots, c_k$$

4) Repeat Step 2 & 3 until convergence

If new centroid and old centroid difference acc. to their dist. is

very small, then they are said to have converged.

Video 8 How to initialize: K-means++

Problem with random initialization is Initialization sensitivity in final clusters & centroids.

1) Repeat K-means multiple times with different initializations and pick the best clustering based on smaller intra cluster and larger inter cluster

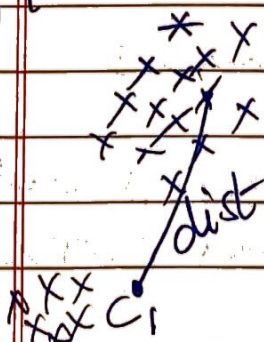
2) K-means++ - instead of using random initialization, do smart initialization which is as follows:

(a) Pick the first centroid randomly $\rightarrow c_1$ from D

(b) $\forall x_i \in D$, create a distribution as:-

$$x_i \rightarrow \text{dist}^2(x_i, \text{nearest centroid})$$

eg



x_1	$\ x_1 - c_1\ ^2 = d_1$
x_2	d_2
$\vdots$	$\vdots$
x_n	d_n

pick a point from $D - c_1$ with prob $\propto d_i$

eg 2

x x x
x x x
x

c_1 x x x
x x x
x x x
x

x x
x x x
x x x
x c_2

In initialization - trying to pick points that are as far as possible from other centroids that are already picked up.

Q why do this probabilistic ally?

Pick a point which has the highest value of $d^2(x_i, \text{nearest centroid})$

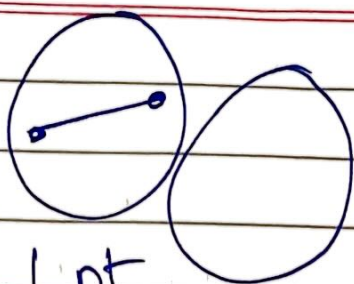
Video 9 Failure cases / Limitations

1. K-means has problems when clusters are of different
a) Size
b) Densities
c) Non-globular shapes

non-convex

2. K-means has problems when the data contains outliers.

Convex set



You reach 1 pt to another in same set.

Overcoming k-means Limitations:

- Use many clusters
 - Find parts of clusters, but need to put together. (not easy)

Video 10 K-Medoids

Problem with K-medoids is that centroids in clusters may not be interpretable.

So, instead of giving real nos. computed using BOW, you give data point, it will be more interpretable.

$$C_i = x_j \in D$$

then it is called K-Medoids.

Algo for K-Medoids:-

Partitioning around ~~med~~ medoids (PAM)

So, first step:-

(1) Initialization - k-means++ -
 Probabilistic method. Pick k pts
 from D .
 Same as k-means

(2) Assignment: Closest medoid
 $x_i \in C_j$ if medoid G_j is
 the closest medoid to x_i ;
 (Same as k-means)

(3) Update / Re-compute:

(a) for each medoid swap with a
 non medoid pt.

M_1 M_2
 $(x_1), x_2, x_3, x_4, x_5, (x_6), x_7, x_8, x_9, x_{10}$
 non medoid non-medoid
 pts pts

(b) If loss decreases, keep the
 swap else undo the swap

(c) If swap is successful
 GOTO (2)

Loss function for k-medoid:-

$$\sum_{i=1}^k \sum_{x \in S_i} \|x - m_j\|^2$$

medoid j

Same example

eg → loss-value $x_1 = M_1, x_2 = M_2 \rightarrow l_1$

⑥ Swap $M_1 = x_2$ x_1 as non-med
 $M_2 = x_6$ ed
pt.

loss-value $M_1 = x_2$ } $\rightarrow l_2$
 $M_2 = x_6$ }

If $l_2 < l_1$

$M_1 = x_2$

$M_2 = x_6$

} swap is successful

else

$M_1 = x_1$

$M_2 = x_6$

keep swapping

Standard K-means is not Kernalizabe

But K-medoids it's much mor
easy to Kernalize.

Video 11 Determining the right K

K-hyperparameter

Determine by:-

1) Domain knowledge

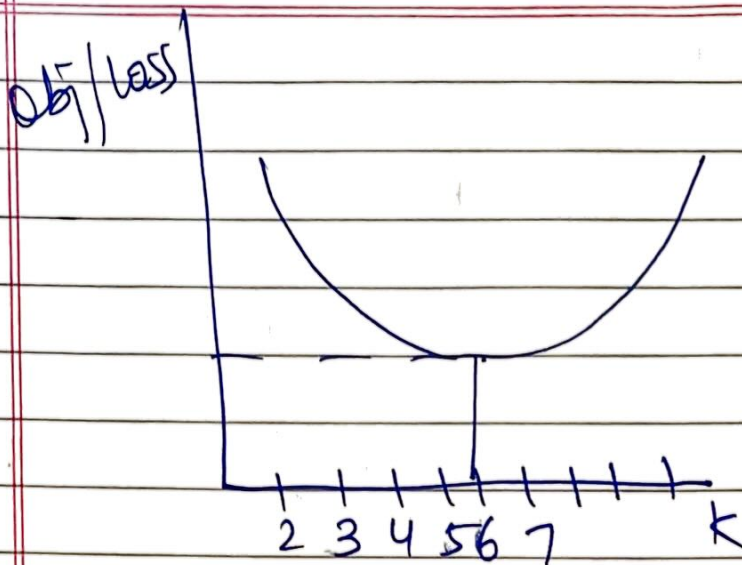
eg- Amazon Fine food Review
+ review
- review

K=2.

Also, use elbow method or
knee method.

loss function of K-means

$$\sum_{i=1}^K \sum_{x \in S_i} \|x - c_i\|^2 \rightarrow \text{minimize}$$



$$K=6$$

best $K=5$ because loss is min.

Video 13 Time and Space Complexity

K-means, Time Complexity

$$O(n k d i)$$

$n = \# \text{pts}$

$k = \# \text{clusters}$

$d = \text{dimension}$

$i = \# \text{iteration}$

Typically, $k \leq 10$
 $i \leq 300$

If n is $100k$ (very large)

$O(nd) \rightarrow \text{linear time complexity}$

Space Complexity

$$O(nd + Kd)$$

$$\approx O(nd) \leftarrow \text{linear}$$

K-means is quite fast.